

Exploring Influences of Appearance and Voice Realism of Virtual Humans on Communication Effectiveness in Social Virtual Reality

Yu Han*

Yu Miao

Hao Sha

Yue Liu† (✉)

Beijing Engineering Research Center of Mixed Reality and Advanced Display
School of Optics and Photonics, Beijing Institute of Technology

ABSTRACT

Virtual humans play a crucial role in elevating user experiences in Virtual Reality (VR). Despite ongoing technological advancements, achieving highly realistic virtual humans with entirely natural behaviors remains a challenge. This paper explores how the appearance and voice realism of virtual humans influence communication effectiveness within social VR scenarios. Our preliminary results indicate the significant impact of alterations in appearance and voice realism on communication effectiveness. We observe that a cross-modal realism mismatch between appearance and voice can hamper effective communication. This research provides valuable insights for designing virtual humans and improving the quality of social communication in VR environments.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI

1 INTRODUCTION

Virtual Reality (VR) has gradually made its way into many aspects of our daily lives, including industries, entertainment, education, and more. Virtual humans, serving as digital representations of individuals, play important roles across various VR applications [4]. Although the recent technological advancements have notably improved the perceived realism of virtual humans, the achievable levels of realism across various aspects of virtual humans may advance at varying paces. Limited by the technological affordability and accessibility, the widespread availability of highly realistic virtual humans with entirely natural behaviors still remains to be a challenge. Evidence suggests that the visual and behavioral realism of virtual humans can significantly influence user perceptions [2] and social interactions [1]. Previous studies have investigated the realism of virtual humans' appearance, voice, and motion on users' perceptions such as emotional response [5], and portrayal of personality [3]. However, a notable gap in the literature exists with regard to communication effectiveness, a pivotal element of social interactions that could have a significant bearing on the efficacy of virtual humans.

In this paper, we explore how the appearance and voice realism of virtual humans influence communication effectiveness in social VR settings. Specifically, we manipulate the realism of a virtual human's appearance and voice, examining the effects of these manipulations on users' retention of the heard speech during interactions with the virtual human. This research could contribute to the design and development of virtual humans within VR scenarios.

2 METHODOLOGY

We altered the realism of a virtual human's appearance and voice, then conducted a user study to examine the effects of these modifi-

cations on the effectiveness of communication with virtual humans.

3 STIMULI CREATION

In investigating appearance and voice realism's impact on social communication in virtual humans, we crafted different realism levels for appearance and voice. Two male avatars, one high realism and the other low realism, were generated from Unity's open-accessible digital human asset¹ and Ready Player Me² online platform (see Fig. 1), both standardized to a height of approximately 1.8 meters. For voice representation, we utilized authentic speech audio recordings sourced from a YouTube video to achieve high realism, and synthesized speech generated from transcribed content through the Microsoft Azure Text-to-Speech (TTS) engine to achieve low realism. The TTS audio files were carefully edited and aligned to match the key timings of the originally recorded real-human speech. We employed Deep Motion³ to generate 3D animations derived from the original speech video obtained from YouTube. These animations were applied to the upper body of both characters, while the lower body received a natural idle animation. Facial animation and lip synchronization were achieved using the Oculus Lipsync Unity plugin. Notably, facial expressions were intentionally omitted to eliminate their potential influence.

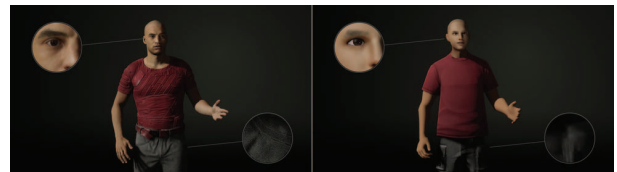


Figure 1: Sample frames of the virtual human with different appearance realism: (left) high realism; (right) low realism.

3.1 Participants and Apparatus

In the experiment, we recruited a total of 40 participants, consisting of 19 males and 21 females, with ages ranging from 22 to 26 years ($M = 23.63$, $SD = 1.46$). The experimental setup comprised an HTC VIVE Pro headset, a 27-inch Philips monitor, and a gaming PC. For the rendering of virtual humans, we utilized Unity (version 2022 LTS) with the High Definition Render Pipeline.

3.2 Experimental Design

The experimental design utilized a 2×2 between-subjects approach, manipulating two independent variables: appearance realism (high vs. low) and voice realism (high vs. low). Consequently, the experiment consisted of a total of four unique trial conditions. The virtual character was positioned in front of the participants, with the

*e-mail: han.yu@outlook.com

† (✉) e-mail: liuyue@bit.edu.cn

¹<https://unity.com/the-heretic>

²<https://readyplayer.me>

³<https://www.deepmotion.com/>

distance between them set to 1.5 m. To mitigate potential environmental influences, the character's surroundings were presented in black, and the character himself was illuminated with stage lighting, as shown in Fig. 1. The experiment comprised two sessions: a practice session and a storytelling session. During the practice session, the character was paired with a 10-second practice audio. The character was animated using a talking idle animation, and both the audio and animation were looped during this practice session. In the storytelling session, the virtual character narrated a brief story about a father and his son, with animations derived from the original YouTube video. The participants' primary task was to listen to the virtual human's narration and subsequently complete a retention test.

3.3 Procedure

In the beginning, participants signed consent forms, followed by wearing VR headsets while standing still. Sound volume adjustments were personalized for clear audio. After participants familiarized themselves with the hardware and the virtual human, the experiment commenced. Upon its conclusion, participants removed the VR headsets. It's important to note that none had prior knowledge of the story. Subsequently, participants undertook a retention test evaluating their recall of the virtual humans' spoken content.

3.4 Measurement

To measure communication effectiveness, we employed a retention test that assessed participants' memory of the information presented in the heard speeches. This test comprised 15 single-choice questions, each related to key information mentioned in the storytelling. It's important to note that effective communication involves a two-way exchange, relying on both the speaker's articulation and the listener's comprehension. In our evaluation, we assume the speaker's verbal expression is adequate. When users listen to a virtual human speak, we evaluate communication through the retention test. Alternatively, when users are speakers, effectiveness depends on the controller of the virtual human—whether it's a real person or an AI. If controlled by AI, appearance or voice realism does not influence it. Yet, if human-controlled, this scenario aligns with the study's focus.

4 RESULTS AND DISCUSSION

IBM SPSS Statistics 25 was used for data analysis. We considered data as statistically significant at the $p < 0.05$ level. To denote significance levels, we used asterisks: *, **, and *** correspond to $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively.

A two-way ANOVA was conducted to analyze the impact of different appearance realism and voice realism conditions on retention test scores. The results revealed a significant main effect for voice realism ($F(1, 36) = 25.04$, $p < 0.001$, $\eta^2_{\text{partial}} = 0.410$), as shown in Fig. 2 (left), implying that variations in the realism of voice had a substantial impact on participants' ability to retain information. However, there was no significant main effect for appearance realism ($F(1, 36) = 0.227$, $p = 0.637$), indicating that varying the realism of appearance alone did not significantly affect participants' retention of the heard speech. Additionally, a significant interaction effect was identified between appearance realism and voice realism ($F(1, 36) = 4.59$, $p = 0.039$, $\eta^2_{\text{partial}} = 0.113$), which suggests that the combined impact of these two factors is more influential than their individual effects. Post hoc comparisons revealed that participants in the "High Appearance Realism and High Voice Realism" condition retained the most information from the spoken speech, suggesting that the seamless integration of lifelike appearance and voice likely contributed to higher levels of attentiveness and focus. In contrast, participants interacting with a virtual human featuring high appearance realism and low voice realism showed the lowest speech retention, indicating that the mismatch between a highly realistic appearance and a less convincing voice could adversely influence participants' information retention.

In summary, our initial findings highlight the importance of maintaining congruence between appearance and voice realism to enhance the retention of spoken information during interactions with virtual humans. This alignment not only fosters effective social communication but also minimizes potential distractions arising from incongruities in appearance and voice realism.

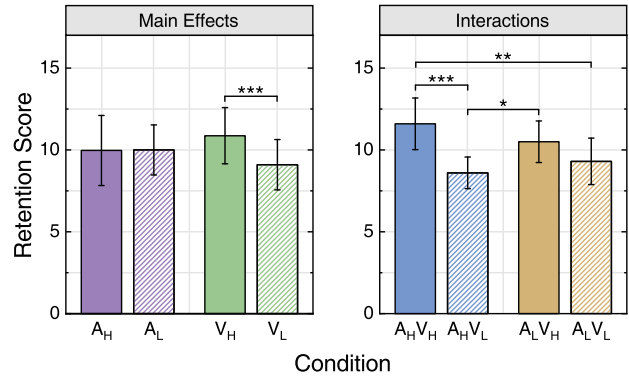


Figure 2: Results of communication effectiveness: (left) main effects; (right) interactions. (A: Appearance; V: Voice; H: High; L: Low)

5 CONCLUSION

In this paper, we investigated how a virtual human's appearance and voice realism influence communication effectiveness during interactions. Our preliminary results suggest that voice realism notably impacted the retention of information, while appearance realism alone did not significantly affect the recall of the heard speech. Additionally, a cross-modal realism mismatch between appearance and voice could potentially hinder communication effectiveness. This research might offer valuable insights into enhancing the design of virtual humans and elevating the quality of social communication within VR environments.

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